LOT Pager ��� 02-BOM-BUYING-LIST

02 · Components Buying List (BOM)

> **Prices are engineering estimates (USD, 2026, qty 1 → qty 100), NOT live quotes.** The sandbox had no vendor access. All links are vendor **search/landing URLs** (stable patterns) — open them, pick an in-stock SKU, confirm datasheet. Re-verify in HW-02.

Tier **A = buy now for EVT/dev-kit**, **B = buy at DVT**, **C = 100-unit run**.

1. Compute & radio

Part	Pick	Why	Qty/unit	Est. $	Link	Tier
MCU (dev)	ESP32-S3-WROOM-1-N16R8 module	Wi-Fi/BLE, DVP camera, PSRAM, vector AI, pre-certified RF	1	4.5	https://www.digikey.com/en/products/result?keywords=ESP32-S3-WROOM-1-N16R8	A
MCU (thin, DVT)	ESP32-S3 bare QFN56 + 16 MB flash + 8 MB PSRAM + chip antenna	Saves ~2.3 mm vs module; needs own RF cert	1	5	https://www.lcsc.com/search?q=ESP32-S3R8	B
Dev board	Espressif ESP32-S3-DevKitC-1	Day-one firmware work	1–3 (once)	15	https://www.digikey.com/en/products/result?keywords=ESP32-S3-DevKitC-1	A

2. Display, camera, input

Part	Pick	Notes	Qty	Est. $	Link	Tier
Screen (EVT)	1.54” 240×240 IPS, ST7789, SPI (≈31 mm active)	Fits 40 mm face; simple, color	1	3.5	https://www.lcsc.com/search?q=1.54%22+ST7789+240x240	A
Screen (alt, low-power)	1.54” e-paper 200×200	Best battery; slow refresh (alert flash via LED)	1	6	https://www.digikey.com/en/products/result?keywords=1.54+e-paper	B
Camera	OV2640 2 MP DVP, 24-pin FPC	ESP32-S3 supported, small	1	3	https://www.lcsc.com/search?q=OV2640	A
Camera (alt)	OV5640 5 MP (AF)	Better image, taller	1	7	https://www.lcsc.com/search?q=OV5640	B
Button	Low-profile tact 0.6 mm travel, ≤1.2	Becomes “Copy”	1	0.4	https://www.digikey.com/en/products/result?keywords=ultra+thin+tactile+switch	A

Part	Pick	Notes	Qty	Est. $	Link	Tier
	mm tall (e.g. Alps SKRP/ C&K PTS) or capacitive-metal dome					
Indicator LED	Camera-active LED 0402 red + RGB 0404 status	Privacy + alerts	2	0.2	https://www.lcsc.com/search?q=0402+LED	A
Haptic (optional pager buzz)	LRA 6 mm flat or piezo buzzer	“Pager” feel	1	1.2	https://www.digikey.com/en/products/result?keywords=flat+vibration+motor+LRA	B

3. Sensors (“AI-grade, off-the-shelf”)

Part	Pick	Measures / AI feature	Est. $	Link	Tier
Weather + air	**Bosch BME688** (T/RH/P + gas, BSEC AI classification)	Weather + air-quality classes	7	https://www.digikey.com/en/products/result?keywords=BME688	A
Precision T/RH	Sensirion SHT45	±0.1 °C reference, far from heat sources	2.5	https://www.digikey.com/en/products/result?keywords=SHT45	B
Ambient light	Vishay VEML7700	Auto screen brightness	1.2	https://www.digikey.com/en/products/result?keywords=VEML7700	A
Motion	**Bosch BMI270** (on-chip gesture/step AI) or ST LSM6DSOX (ML core)	Tap/wake/orientation	2	https://www.digikey.com/en/products/result?keywords=BMI270	A
UV (optional)	LTR-390UV-01	Weather “UV index”	1.5	https://www.digikey.com/en/products/result?keywords=LTR-390UV	C
On-device AI	TensorFlow Lite Micro / ESP-DL on ESP32-S3 (software)	Person/motion detect on camera	0	https://github.com/espressif/esp-dl	A

Thermal note: ESP32-S3 + charging heat biases the T/RH sensor — place sensors at board edge on a thermal slot, firmware offset calibration.

4. Power & wireless charging

Part	Pick	Notes	Est. $	Link	Tier
Battery	LiPo **3.0 mm** thick (e.g. 303040 ≈ 3×30×40 mm, ~280 mAh; or 302530 ~180	Thickness-limited	3.5	https://www.lcsc.com/search?q=303040+lipo	A

Part	Pick	Notes	Est. $	Link	Tier
	mAh), PCM + UN38.3 doc				
Qi receiver IC	TI **BQ51013B** (5 V out) or Cypress/Infineon/CPS equivalent	Qi 1.2 BPP	2.2	https://www.digikey.com/en/products/result?keywords=BQ51013B	A
Qi RX coil	Ultra-thin ≈ 0.5–0.8 mm, ≤ 30 mm dia, Ni-Zn ferrite shield, ~ 10 µH (e.g. Würth 760308xxxx series or Chinese FPC coil)	Needs window (R2)	3	https://www.digikey.com/en/products/result?keywords=wireless+charging+receiver+coil	A
Li-ion charger	TI **BQ25185** (≤1 A, power-path, tiny)	USB-C + Qi 5 V in	1	https://www.digikey.com/en/products/result?keywords=BQ25185	A
Fuel gauge	Analog/Maxim **MAX17048**	% battery on screen/site	1.5	https://www.digikey.com/en/products/result?keywords=MAX17048	A
3.3 V buck	TPS62840 (60 nA Iq)	Battery life	1	https://www.digikey.com/en/products/result?keywords=TPS62840	A
USB-C (service/charge)	Midmount low-profile 2-pin power-only, or **pogo pads** on seam	USB-C is 3.2 mm → pogo pads preferred	0.5	https://www.digikey.com/en/products/result?keywords=pogo+pin+pads	B
Charging pad (for desk)	Any Qi 5 W pad for EVT; custom LOT pad at DVT		12	https://www.digikey.com/en/products/result?keywords=qi+wireless+charging+transmitter+module	A

5. PCB, mechanics (all PCBWay)

Item	Spec	Est.	Link	Tier
PCB EVT	4-layer, 0.8 mm, ENIG, 38×38 mm, 5–10 pcs; (option: rigid-flex for camera/screen)	$60–150 / batch	https://www.pcbway.com/orderonline.aspx	A
SMT assembly	PCBWay turnkey, fine pitch, incl. stencil	$150–350 EVT	https://www.pcbway.com/pcb-assembly.html	A
Steel body (2 parts)	**316L** CNC, back: 0.6–1.0 mm plate mirror-polished (#8); front: bezel with screen/camera/button openings	$20–35/part @100	https://www.pcbway.com/cnc-machining-quote.html	B
Dielectric windows	Alumina/zirconia ceramic or sapphire disc (Qi), PEEK/PC ring (RF seam)	$2–6	https://www.pcbway.com/cnc-machining-quote.html	B
Cover glass	0.5 mm chemically-strengthened glass over screen, AR	$1.5	https://www.lcsc.com/search?q=cover+glass	B
Camera lens window	Sapphire/glass 4 mm	$0.8	(as above)	B

Item	Spec	Est.	Link	Tier
EVT shells (before steel)	3D print / CNC aluminium	$40–100	https://www.pcbway.com/rapid-prototyping/	A

6. Unit-cost roll-up (estimates, qty 100)

Group	$/unit (low–high)
Compute & radio	5–9
Display + camera + button + LEDs	8–15
Sensors	10–16
Power & Qi	12–18
PCB + SMT (100 run)	9–16
Steel ×2 + windows + glass	45–85
Packaging, test fixture amortization, assembly labor	8–15
Total	**≈ 97–175**

100-unit run ≈ $10 k–17.5 k + NRE (stencil, test jig, CNC programming ≈ $1–2.5 k) + certification ($6–15 k; module-based design is cheaper).

7. Procurement order of operations

1. **This week (Tier A, ≈ $150–250):** 2× DevKitC-1, 2× 1.54” TFT, 2× OV2640, BME688 breakout, BMI270 breakout, VEML7700 breakout, Qi pad + RX kit, 2× 303040 LiPo, buttons. → proves firmware & site loop.
2. **After P3 Qi/RF test:** release Tier B (CNC steel prototypes + windows).
3. **After DVT sign-off:** Tier C and PCBWay 100-unit order.